

```
In [100... import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np
```

```
In [101... pd.set_option('display.max_columns',None)
sns.set_style('whitegrid')
```

```
df=pd.read_csv('german_credit_data.csv')
print(df)
```

	Unnamed: 0	Age	Sex	Job	Housing	Saving accounts	Checking account \
0	0	67	male	2	own	NaN	little
1	1	22	female	2	own	little	moderate
2	2	49	male	1	own	little	NaN
3	3	45	male	2	free	little	little
4	4	53	male	2	free	little	little
..	...	...	...	...	...	...	...
995	995	31	female	1	own	little	NaN
996	996	40	male	3	own	little	little
997	997	38	male	2	own	little	NaN
998	998	23	male	2	free	little	little
999	999	27	male	2	own	moderate	moderate

	Credit amount	Duration	Purpose	Risk
0	1169	6	radio/TV	good
1	5951	48	radio/TV	bad
2	2096	12	education	good
3	7882	42	furniture/equipment	good
4	4870	24	car	bad
..	...	...	...	...
995	1736	12	furniture/equipment	good
996	3857	30	car	good
997	804	12	radio/TV	good
998	1845	45	radio/TV	bad
999	4576	45	car	good

[1000 rows x 11 columns]

```
In [102... print(df.head())

print(df['Risk'].value_counts())
```

	Unnamed: 0	Age	Sex	Job	Housing	Saving accounts	Checking account	\
0	0	67	male	2	own	NaN	little	
1	1	22	female	2	own	little	moderate	
2	2	49	male	1	own	little	NaN	
3	3	45	male	2	free	little	little	
4	4	53	male	2	free	little	little	

	Credit amount	Duration	Purpose	Risk
0	1169	6	radio/TV	good
1	5951	48	radio/TV	bad
2	2096	12	education	good
3	7882	42	furniture/equipment	good
4	4870	24	car	bad

Risk

good 700

bad 300

Name: count, dtype: int64

In [103... df.shape

Out[103... (1000, 11)

In [104... df.info()

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 1000 entries, 0 to 999

Data columns (total 11 columns):

#	Column	Non-Null Count	Dtype
0	Unnamed: 0	1000 non-null	int64
1	Age	1000 non-null	int64
2	Sex	1000 non-null	object
3	Job	1000 non-null	int64
4	Housing	1000 non-null	object
5	Saving accounts	817 non-null	object
6	Checking account	606 non-null	object
7	Credit amount	1000 non-null	int64
8	Duration	1000 non-null	int64
9	Purpose	1000 non-null	object
10	Risk	1000 non-null	object

dtypes: int64(5), object(6)

memory usage: 86.1+ KB

In [105... df.describe(include='all').T

Out[105...

	count	unique	top	freq	mean	std	min	25%	50%
Unnamed: 0	1000.0	NaN	NaN	NaN	499.5	288.819436	0.0	249.75	499.5
Age	1000.0	NaN	NaN	NaN	35.546	11.375469	19.0	27.0	33.0
Sex	1000	2	male	690	NaN	NaN	NaN	NaN	NaN
Job	1000.0	NaN	NaN	NaN	1.904	0.653614	0.0	2.0	2.0
Housing	1000	3	own	713	NaN	NaN	NaN	NaN	NaN
Saving accounts	817	4	little	603	NaN	NaN	NaN	NaN	NaN
Checking account	606	3	little	274	NaN	NaN	NaN	NaN	NaN
Credit amount	1000.0	NaN	NaN	NaN	3271.258	2822.736876	250.0	1365.5	2319.5
Duration	1000.0	NaN	NaN	NaN	20.903	12.058814	4.0	12.0	18.0
Purpose	1000	8	car	337	NaN	NaN	NaN	NaN	NaN
Risk	1000	2	good	700	NaN	NaN	NaN	NaN	NaN

In [106...

df.isna().sum()

Out[106...

```

Unnamed: 0      0
Age             0
Sex             0
Job             0
Housing         0
Saving accounts 183
Checking account 394
Credit amount   0
Duration         0
Purpose          0
Risk            0
dtype: int64

```

In [107...

```

df=df.dropna().reset_index(drop=True)
df

```

Out[107...

	Unnamed: 0	Age	Sex	Job	Housing	Saving accounts	Checking account	Credit amount	Duration
0	1	22	female	2	own	little	moderate	5951	48
1	3	45	male	2	free	little	little	7882	42
2	4	53	male	2	free	little	little	4870	24
3	7	35	male	3	rent	little	moderate	6948	36
4	9	28	male	3	own	little	moderate	5234	30
...	...	...	...	...	...	...	...	...	...
517	989	48	male	1	own	little	moderate	1743	24
518	993	30	male	3	own	little	little	3959	36
519	996	40	male	3	own	little	little	3857	30
520	998	23	male	2	free	little	little	1845	45
521	999	27	male	2	own	moderate	moderate	4576	45

522 rows × 11 columns



In [108...

df.columns

Out[108...

```
Index(['Unnamed: 0', 'Age', 'Sex', 'Job', 'Housing', 'Saving accounts',
      'Checking account', 'Credit amount', 'Duration', 'Purpose', 'Risk'],
      dtype='object')
```

In [109...

df.drop(columns='Unnamed: 0',inplace=True)

In [110...

df.columns

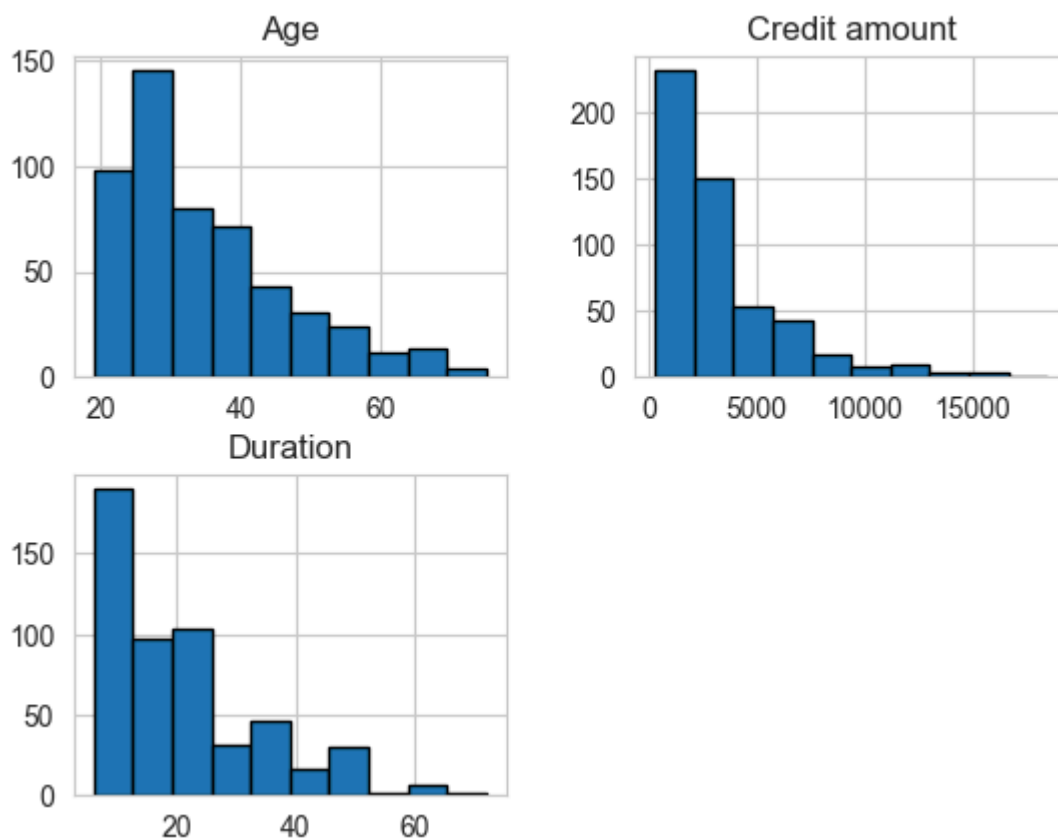
Out[110...

```
Index(['Age', 'Sex', 'Job', 'Housing', 'Saving accounts', 'Checking account',
      'Credit amount', 'Duration', 'Purpose', 'Risk'],
      dtype='object')
```

In [111...

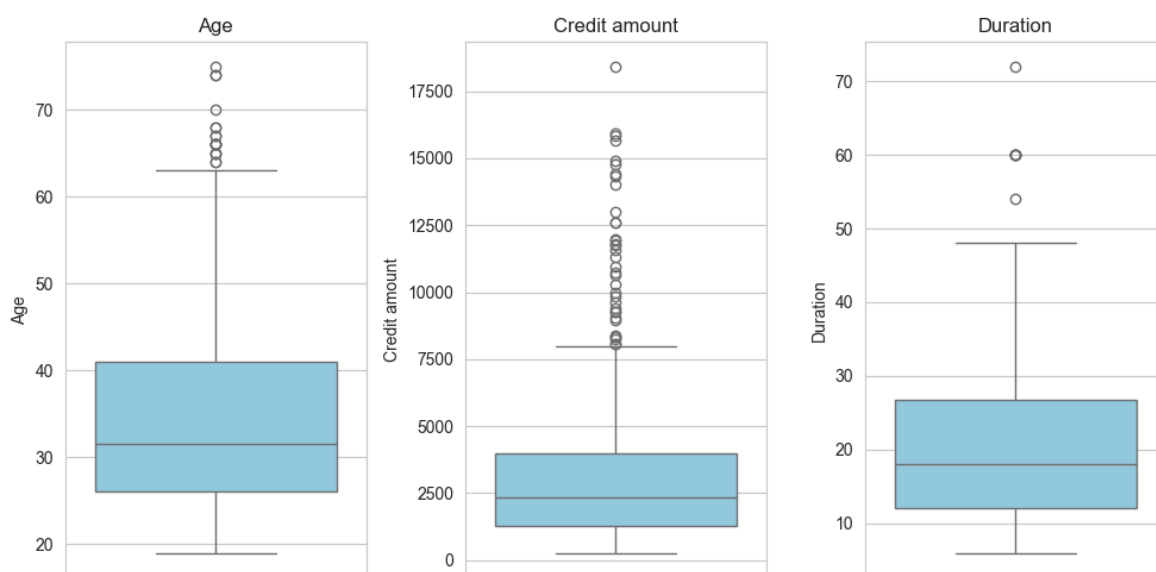
```
df[['Age','Credit amount','Duration']].hist(bins=10,edgecolor='black')
plt.suptitle('distribution of numerical features',fontsize=14)
plt.show()
```

distribution of numerical features



```
In [112... plt.figure(figsize=(10,5))
for i,col in enumerate(['Age','Credit amount','Duration']):
    plt.subplot(1,3,i+1)
    sns.boxplot(y=df[col],color='skyblue')
    plt.title(col)

plt.tight_layout()
plt.show()
```



```
In [113... categorical_cols=['Sex','Job','Housing','Saving accounts','Checking account','Pu
```

```
In [114... plt.figure(figsize=(15,10))
for i,col in enumerate(categorical_cols):
    plt.subplot(3,3,i+1)
    sns.countplot(data=df,x=col,palette='Set2',order=df[col].value_counts().inde
    plt.title(f"distribution of {col}")
    plt.xticks(rotation=45 )

plt.tight_layout()
plt.show()
```

```
C:\Users\srish\AppData\Local\Temp\ipykernel_6588\3849260992.py:4: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v
0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effe
ct.

sns.countplot(data=df,x=col,palette='Set2',order=df[col].value_counts().index,l
egend=False)
C:\Users\srish\AppData\Local\Temp\ipykernel_6588\3849260992.py:4: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v
0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effe
ct.

sns.countplot(data=df,x=col,palette='Set2',order=df[col].value_counts().index,l
egend=False)
C:\Users\srish\AppData\Local\Temp\ipykernel_6588\3849260992.py:4: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v
0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effe
ct.

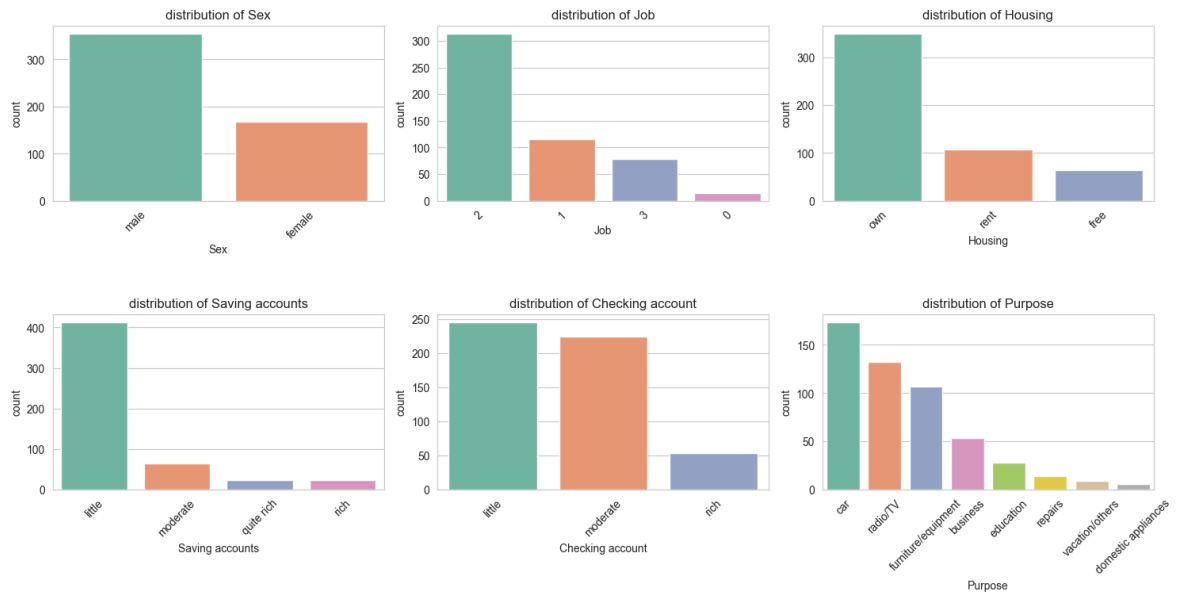
sns.countplot(data=df,x=col,palette='Set2',order=df[col].value_counts().index,l
egend=False)
C:\Users\srish\AppData\Local\Temp\ipykernel_6588\3849260992.py:4: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v
0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effe
ct.

sns.countplot(data=df,x=col,palette='Set2',order=df[col].value_counts().index,l
egend=False)
C:\Users\srish\AppData\Local\Temp\ipykernel_6588\3849260992.py:4: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v
0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effe
ct.

sns.countplot(data=df,x=col,palette='Set2',order=df[col].value_counts().index,l
legend=False)
```

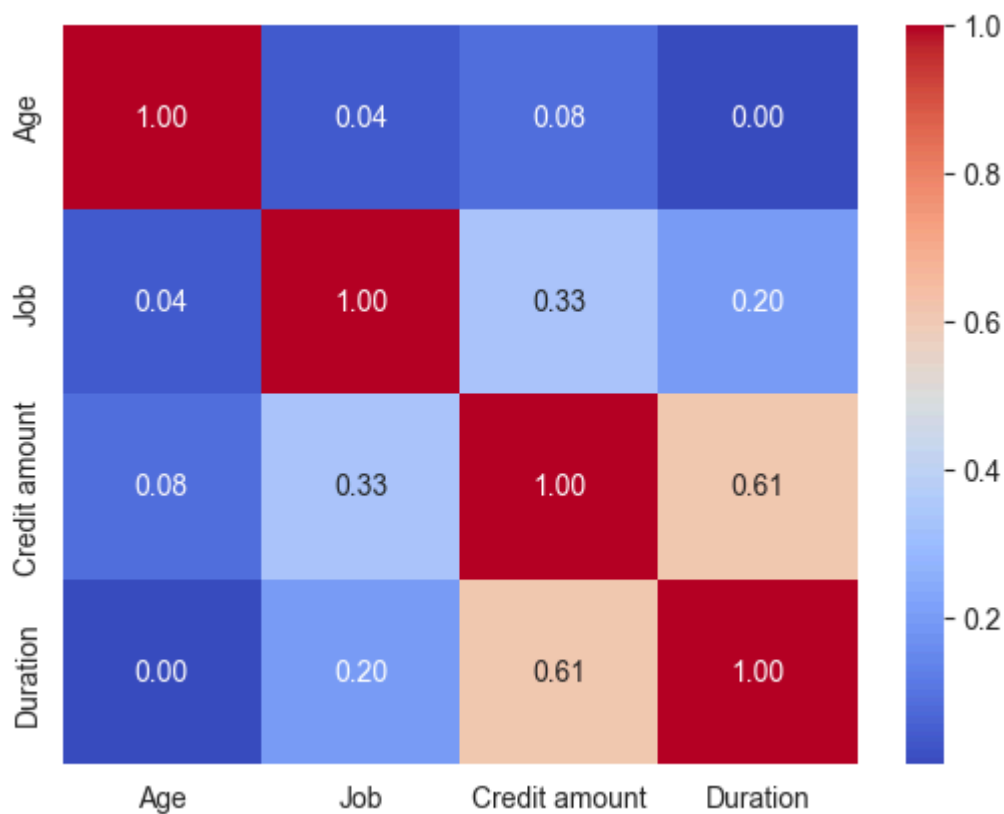


```
In [115]: corr=df[['Age','Job','Credit amount','Duration']].corr()
          corr
```

```
Out[115]:
```

	Age	Job	Credit amount	Duration
Age	1.000000	0.039771	0.082014	0.001549
Job	0.039771	1.000000	0.334721	0.200794
Credit amount	0.082014	0.334721	1.000000	0.613298
Duration	0.001549	0.200794	0.613298	1.000000

```
In [116]: sns.heatmap(corr,annot=True,cmap='coolwarm',fmt='.2f')
          plt.show()
```

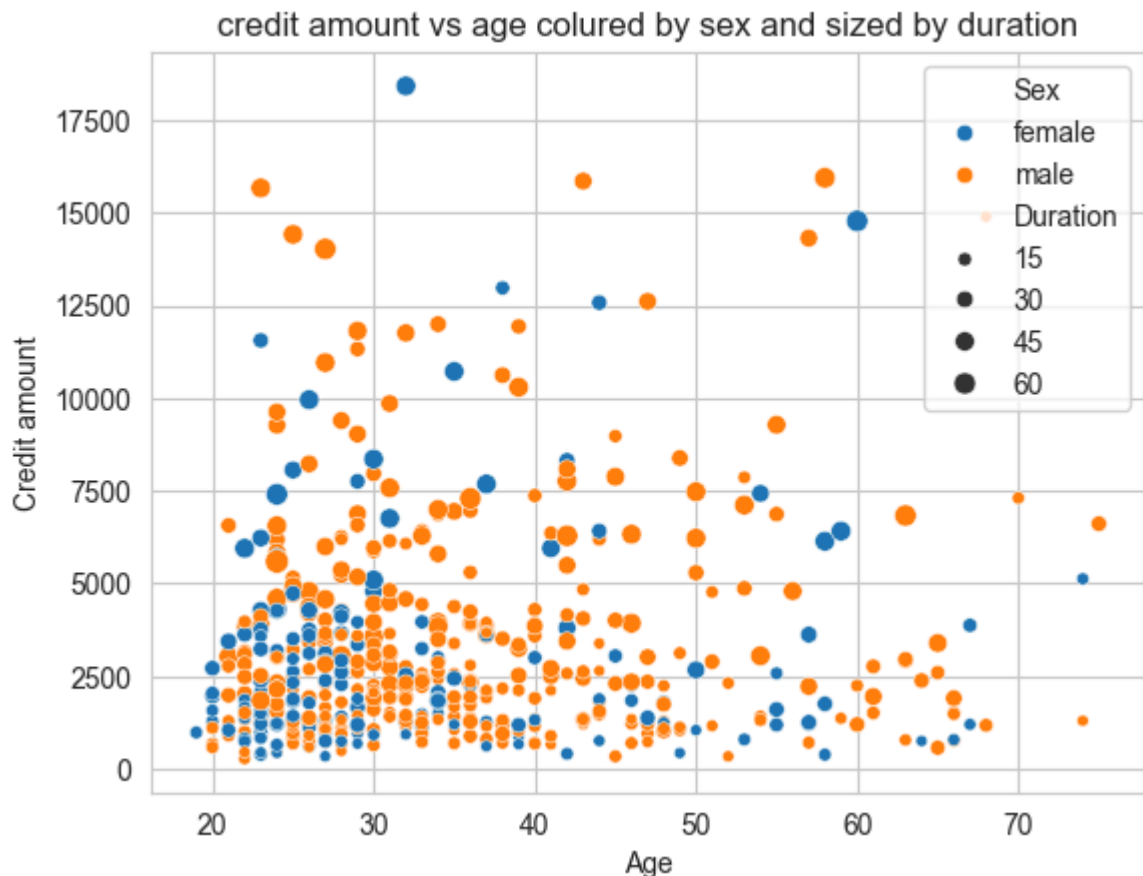


```
In [117... df.groupby('Job')['Credit amount'].mean()

df.groupby('Sex')['Credit amount'].mean()
```

```
Out[117... Sex
female    2937.202381
male      3440.833333
Name: Credit amount, dtype: float64
```

```
In [118... sns.scatterplot(data=df,x='Age',y='Credit amount',hue='Sex',size='Duration')
plt.title(('credit amount vs age colored by sex and sized by duration'))
plt.show()
```

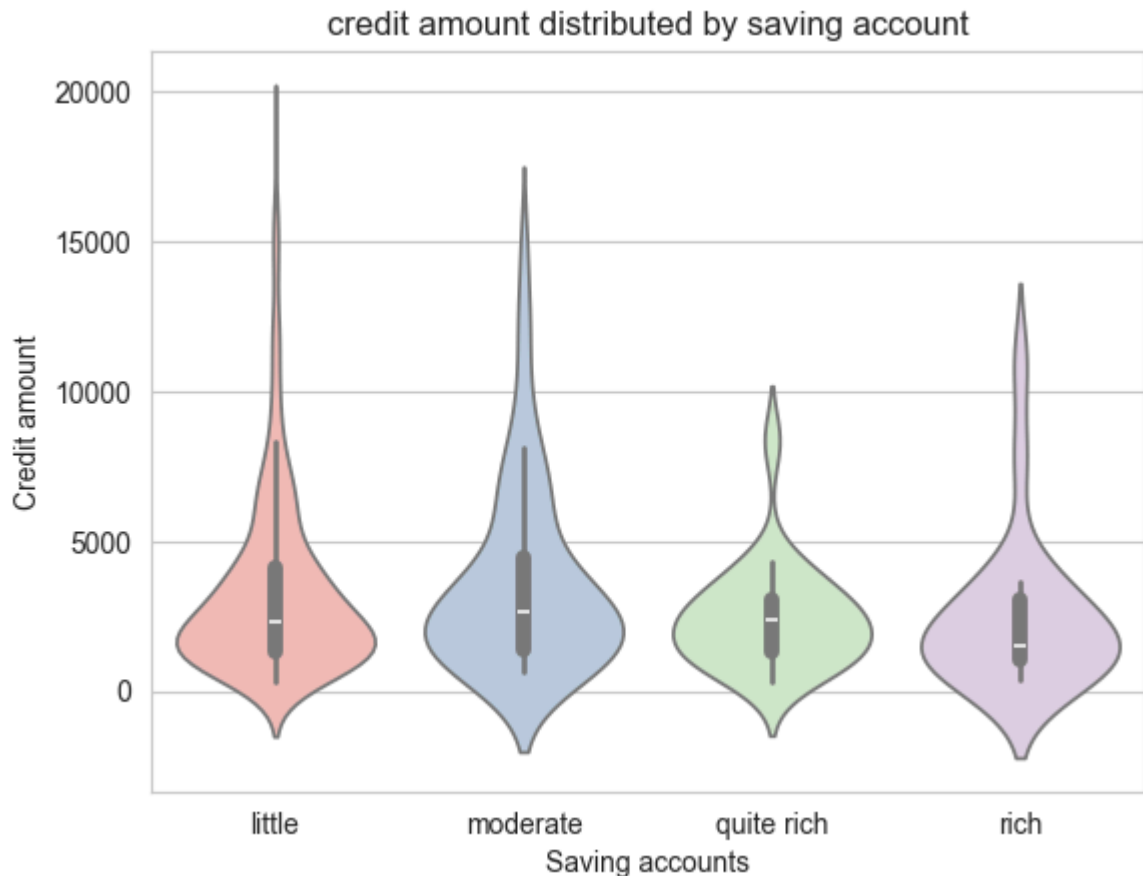


```
In [119... sns.violinplot(data=df,x='Saving accounts',y='Credit amount',palette='Pastel1')
plt.title('credit amount distributed by saving account')
plt.show()
```

C:\Users\srish\AppData\Local\Temp\ipykernel_6588\3931359287.py:1: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v 0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.violinplot(data=df,x='Saving accounts',y='Credit amount',palette='Pastel1')
```



```
In [120... plt.figure(figsize=(10,5))
for i,col in enumerate(['Age','Credit amount','Duration']):
    plt.subplot(1,3,i+1)
    sns.boxplot(data=df,x='Risk',y=col,palette='Pastel2')
    plt.title(f"{col} by risk")

plt.tight_layout()
plt.show()
```

C:\Users\srish\AppData\Local\Temp\ipykernel_6588\3894981837.py:4: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v 0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.boxplot(data=df,x='Risk',y=col,palette='Pastel2')
```

C:\Users\srish\AppData\Local\Temp\ipykernel_6588\3894981837.py:4: FutureWarning:

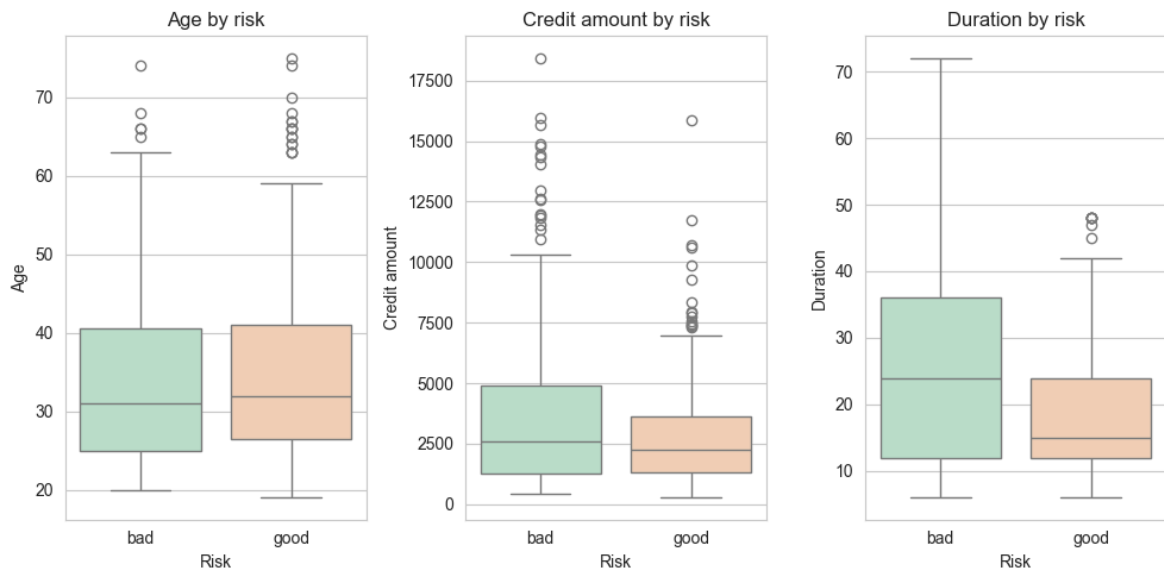
Passing `palette` without assigning `hue` is deprecated and will be removed in v 0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.boxplot(data=df,x='Risk',y=col,palette='Pastel2')
```

C:\Users\srish\AppData\Local\Temp\ipykernel_6588\3894981837.py:4: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v 0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.boxplot(data=df,x='Risk',y=col,palette='Pastel2')
```

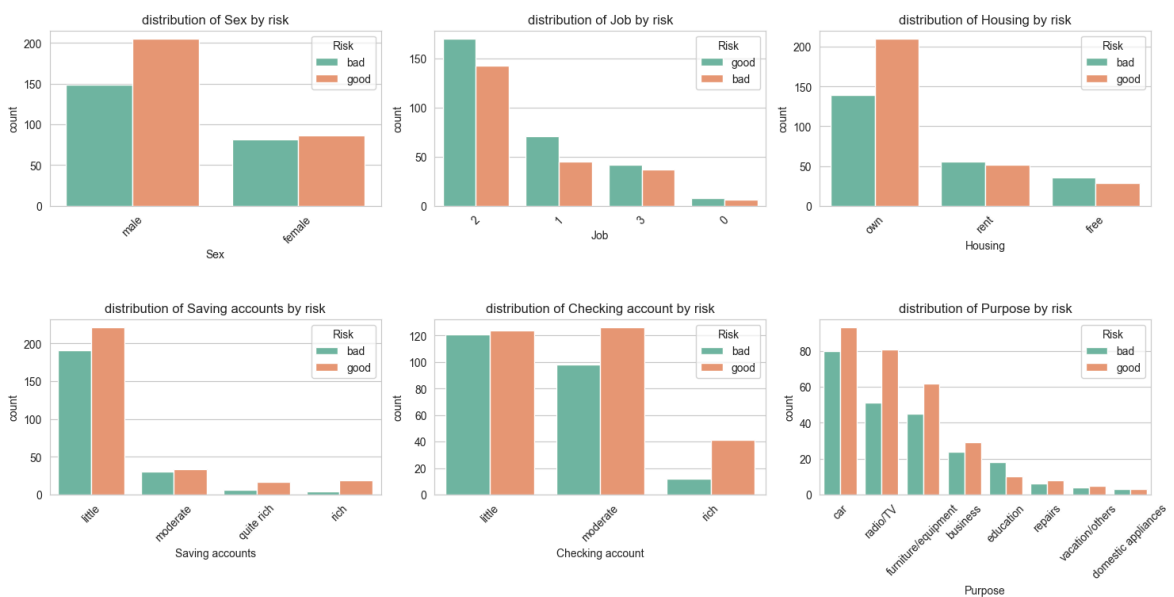


```
In [121... df.groupby('Risk')[['Age', 'Credit amount', 'Duration']].mean()
```

```
Out[121...      Age  Credit amount  Duration
Risk
bad   34.147186    3881.090909  25.445887
good  35.477663    2800.594502  18.079038
```

```
In [122... plt.figure(figsize=(15,10))
for i,col in enumerate(categorical_cols):
    plt.subplot(3,3,i+1)
    sns.countplot(data=df,x=col,hue='Risk',palette='Set2',order=df[col].value_co
    plt.title(f"distribution of {col} by risk")
    plt.xticks(rotation=45 )

plt.tight_layout()
plt.show()
```



```
In [123... features=['Age', 'Sex', 'Job', 'Housing', 'Saving accounts', 'Checking account', 'Credit amount', 'Duration', 'Risk']
target='Risk'
```

```
In [124... df_model=df[features+[target]].copy()
df_model.head()
```

```
Out[124... 
```

	Age	Sex	Job	Housing	Saving accounts	Checking account	Credit amount	Duration	Risk
0	22	female	2	own	little	moderate	5951	48	bad
1	45	male	2	free	little	little	7882	42	good
2	53	male	2	free	little	little	4870	24	bad
3	35	male	3	rent	little	moderate	6948	36	good
4	28	male	3	own	little	moderate	5234	30	bad

```
In [125... import joblib
from sklearn.preprocessing import LabelEncoder
```

```
In [126... cat_cols=df_model.select_dtypes(include='object').columns.drop('Risk')
```

```
In [127... le_dict={}
```

```
In [128... for col in cat_cols:
    le=LabelEncoder()
    df_model[col]=le.fit_transform(df_model[col])
    le_dict[col]=le
    joblib.dump(le, f"{col}_encoder.pkl")
```

```
In [129... le_target=LabelEncoder()
```

```
In [130... target
```

```
Out[130... 'Risk'
```

```
In [131... df_model[target]=le_target.fit_transform(df_model[target])
```

```
In [132... df_model[target].value_counts()
```

```
Out[132... Risk
1    291
0    231
Name: count, dtype: int64
```

```
In [133... joblib.dump(le_target, "target_encoder.pkl")
```

```
Out[133... ['target_encoder.pkl']
```

```
In [134... df_model.head()
```

Out[134...

	Age	Sex	Job	Housing	Saving accounts	Checking account	Credit amount	Duration	Risk
0	22	0	2	1	0	1	5951	48	0
1	45	1	2	0	0	0	7882	42	1
2	53	1	2	0	0	0	4870	24	0
3	35	1	3	2	0	1	6948	36	1
4	28	1	3	1	0	1	5234	30	0

In [135...

```
from sklearn.model_selection import train_test_split
```

In [136...

```
X=df_model.drop(target,axis=1)
y=df_model[target]
```

In [137...

```
X
```

Out[137...

	Age	Sex	Job	Housing	Saving accounts	Checking account	Credit amount	Duration
0	22	0	2	1	0	1	5951	48
1	45	1	2	0	0	0	7882	42
2	53	1	2	0	0	0	4870	24
3	35	1	3	2	0	1	6948	36
4	28	1	3	1	0	1	5234	30
...	...	...	...	...	...	...	...	...
517	48	1	1	1	0	1	1743	24
518	30	1	3	1	0	0	3959	36
519	40	1	3	1	0	0	3857	30
520	23	1	2	0	0	0	1845	45
521	27	1	2	1	1	1	4576	45

522 rows × 8 columns

In [138...

```
x_train,x_test,y_train,y_test=train_test_split(X,y,test_size=0.2,stratify=y,rand
```

In [139...

```
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier,ExtraTreesClassifier
from xgboost import XGBClassifier
from sklearn.metrics import accuracy_score
from sklearn.model_selection import GridSearchCV
```

In [140...

```
def train_model(model,param_grid,x_train,y_train,x_test,y_test):
    grid=GridSearchCV(model,param_grid,cv=5,scoring="accuracy",n_jobs=-1)
    grid.fit(x_train,y_train)
    best_model=grid.best_estimator_
    y_pred=best_model.predict(x_test)
```

```
acc=accuracy_score(y_test,y_pred)
return best_model,acc,grid.best_params_
```

```
In [141... dt=DecisionTreeClassifier(random_state=1,class_weight='balanced')
dt_param_grid={'max_depth':[3,5,7,10,None], 'min_samples_split':[2,5,10], 'min_sam
```

```
In [142... best_dt,acc_dt,params_dt=train_model(dt,dt_param_grid,x_train,y_train,x_test,y_t
```

```
In [143... print("decision_tree_accuracy",acc_dt)
```

decision_tree_accuracy 0.580952380952381

```
In [144... print("decision_tree_model",best_dt)
```

decision_tree_model DecisionTreeClassifier(class_weight='balanced', max_depth=5, random_state=1)

```
In [145... print("best_param",params_dt)
```

best_param {'max_depth': 5, 'min_samples_leaf': 1, 'min_samples_split': 2}

```
In [146... rf=RandomForestClassifier(random_state=1,class_weight='balanced',n_jobs=-1)
rf_param_grid={'n_estimators':[100,200], 'max_depth':[5,7,10,None], 'min_samples_s
```

```
In [147... best_rf,acc_rf,param_rf=train_model(rf,rf_param_grid,x_train,y_train,x_test,y_te
```

```
In [148... print("random_forest_accuracy",acc_rf)
```

random_forest_accuracy 0.6190476190476191

```
In [149... print("random_forest_model",best_rf)
```

random_forest_model RandomForestClassifier(class_weight='balanced', min_samples_leaf=2, min_samples_split=10, n_jobs=-1, random_state=1)

```
In [150... print("best params",param_rf)
```

best params {'max_depth': None, 'min_samples_leaf': 2, 'min_samples_split': 10, 'n_estimators': 100}

```
In [151... et=ExtraTreesClassifier(random_state=1,class_weight='balanced',n_jobs=-1)
et_param_grid={'n_estimators':[100,200], 'max_depth':[5,7,10,None], 'min_samples_s
```

```
In [152... best_et,acc_et,params_et=train_model(et,et_param_grid,x_train,y_train,x_test,y_t
```

```
In [153... print("extra trees accuracy",acc_et)
```

extra trees accuracy 0.6476190476190476

```
In [154... print('best_params',params_et)
```

best_params {'max_depth': 10, 'min_samples_leaf': 2, 'min_samples_split': 5, 'n_estimators': 100}

```
In [155... xgb=XGBClassifier(random_state=1,scale_pos_weight=(y_train==0).sum()/(y_train==1
```

```
In [156... xgb_param_grid={
    "n_estimators":[100,200],
    "max_depth":[3,5,7],
    "learning_rate":[0.01,0.1,0.2],
```

```
"subsample": [0.7, 1],
"colsample_bytree": [0.7, 1]
}
```

```
In [157... best_xgb,acc_xgb,params_xgb=train_model(xgb,xgb_param_grid,x_train,y_train,x_test)
```

```
c:\Users\srish\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\xgboost
\training.py:200: UserWarning: [17:44:30] WARNING: C:\actions-runner\_work\xgboos
t\xgboost\src\learner.cc:793:
```

Parameters: { "use_label_encoder" } are not used.

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
In [158... print("XGB accuracy",acc_xgb)
```

XGB accuracy 0.6666666666666666

```
In [159... best_et.predict(x_test)
```

```
Out[159... array([1, 1, 1, 1, 1, 0, 0, 0, 0, 1, 0, 0, 1, 1, 1, 1, 0, 0, 0, 1, 1, 0,
        1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 1, 1, 1, 0, 0, 1,
        1, 1, 1, 0, 0, 0, 1, 1, 0, 0, 1, 0, 1, 0, 0, 0, 0, 1, 1, 1, 1, 0,
        0, 1, 1, 1, 1, 1, 0, 0, 1, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 0, 1,
        0, 0, 1, 1, 1, 0, 0, 1, 0, 0, 1, 0, 1, 1, 1, 0, 1])
```

```
In [160... joblib.dump(best_et, "extra_tress_credit_model.pkl")
```

```
Out[160]: ['extra_tress_credit_model.pkl']
```

```
In [161... %pip install shap
```

Collecting shap

Downloading shap-0.52.0-cp312-abi3-win_amd64.whl.metadata (26 kB)

Requirement already satisfied: numpy>=2 in c:\Users\srish\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from shap) (2.4.1)

Requirement already satisfied: scipy in c:\Users\srish\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from shap) (1.17.0)

Requirement already satisfied: scikit-learn in c:\Users\srish\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from shap) (1.8.0)

Requirement already satisfied: pandas in c:\Users\srish\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from shap) (2.3.3)

Collecting tqdm>=4.27.0 (from shap)

Downloading tqdm-4.69.0-py3-none-any.whl.metadata (57 kB)

Requirement already satisfied: packaging>20.9 in c:\Users\srish\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from shap) (25.0)

Collecting slicer==0.0.8 (from shap)

Downloading slicer-0.0.8-py3-none-any.whl.metadata (4.0 kB)

Collecting numba (from shap)

Downloading numba-0.66.0-cp314-cp314-win_amd64.whl.metadata (2.8 kB)

Collecting llvmlite (from shap)

Downloading llvmlite-0.48.0-cp314-cp314-win_amd64.whl.metadata (5.2 kB)

Collecting cloudpickle (from shap)

Downloading cloudpickle-3.1.2-py3-none-any.whl.metadata (7.1 kB)

Requirement already satisfied: colorama in c:\Users\srish\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from tqdm>=4.27.0->shap) (0.4.6)

Requirement already satisfied: python-dateutil>=2.8.2 in c:\Users\srish\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from pandas->shap) (2.9.0.post0)

Requirement already satisfied: pytz>=2020.1 in c:\Users\srish\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from pandas->shap) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in c:\Users\srish\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from pandas->shap) (2025.3)

Requirement already satisfied: six>=1.5 in c:\Users\srish\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from python-dateutil>=2.8.2->pandas->shap) (1.17.0)

Requirement already satisfied: joblib>=1.3.0 in c:\Users\srish\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from scikit-learn->shap) (1.5.3)

Requirement already satisfied: threadpoolctl>=3.2.0 in c:\Users\srish\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from scikit-learn->shap) (3.6.0)

Downloading shap-0.52.0-cp312-abi3-win_amd64.whl (499 kB)

Downloading slicer-0.0.8-py3-none-any.whl (15 kB)

Downloading tqdm-4.69.0-py3-none-any.whl (676 kB)

----- 0.0/676.7 kB ? eta -:--:--

----- 676.7/676.7 kB 7.8 MB/s 0:00:00

Downloading cloudpickle-3.1.2-py3-none-any.whl (22 kB)

Downloading llvmlite-0.48.0-cp314-cp314-win_amd64.whl (43.0 MB)

----- 0.0/43.0 MB ? eta -:--:--

----- 5.2/43.0 MB 26.2 MB/s eta 0:00:02

----- 10.7/43.0 MB 26.7 MB/s eta 0:00:02

----- 16.5/43.0 MB 26.6 MB/s eta 0:00:01

----- 22.5/43.0 MB 26.9 MB/s eta 0:00:01

----- 28.0/43.0 MB 26.8 MB/s eta 0:00:01

----- 33.6/43.0 MB 26.9 MB/s eta 0:00:01

----- 39.6/43.0 MB 26.9 MB/s eta 0:00:01

----- 42.7/43.0 MB 26.8 MB/s eta 0:00:01

----- 42.7/43.0 MB 26.8 MB/s eta 0:00:01

----- 43.0/43.0 MB 21.9 MB/s 0:00:01

Downloading numba-0.66.0-cp314-cp314-win_amd64.whl (2.8 MB)

----- 0.0/2.8 MB ? eta -:--:--

----- 2.8/2.8 MB 21.6 MB/s 0:00:00

Installing collected packages: tqdm, slicer, llvmlite, cloudpickle, numba, shap

17/28

[illegible]


```
In [162... import shap
import matplotlib.pyplot as plt
```

```
c:\Users\srish\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\tqdm\auto.py:21: TqdmWarning: IProgress not found. Please update jupyter and ipywidgets.
See https://ipywidgets.readthedocs.io/en/stable/user_install.html
    from .autonotebook import tqdm as notebook_tqdm
```

```
In [183... explainer = shap.Explainer(best_et)
```

```
In [184... explainer
```

```
Out[184... <shap.explainers._tree.TreeExplainer at 0x21a9d886ad0>
```

```
In [185... shap_values = explainer(x_test)
shap_values
```

```

Out[185... .values =
array([[ 0.00439902, -0.00439902],
       [-0.04521615,  0.04521615],
       [-0.02042354,  0.02042354],
       ...,
       [ 0.02640314, -0.02640314],
       [-0.02429964,  0.02429964],
       [-0.0795931 ,  0.0795931 ]],

       [[-0.01606624,  0.01606624],
        [-0.02836201,  0.02836201],
        [-0.05473448,  0.05473448],
        ...,
        [ 0.00906109, -0.00906109],
        [-0.0157617 ,  0.0157617 ],
        [-0.03054655,  0.03054655]],

       [[ 0.00293275, -0.00293275],
        [-0.05562514,  0.05562514],
        [-0.00377769,  0.00377769],
        ...,
        [-0.00547076,  0.00547076],
        [ 0.00930054, -0.00930054],
        [-0.07390892,  0.07390892]],

       ...,

       [[-0.05335445,  0.05335445],
        [ 0.05291754, -0.05291754],
        [ 0.03483129, -0.03483129],
        ...,
        [ 0.01598851, -0.01598851],
        [-0.01073653,  0.01073653],
        [-0.04489548,  0.04489548]],

       [[ 0.00838806, -0.00838806],
        [-0.03412412,  0.03412412],
        [ 0.01487223, -0.01487223],
        ...,
        [ 0.02688523, -0.02688523],
        [-0.01987561,  0.01987561],
        [ 0.08198507, -0.08198507]],

       [[ 0.00344296, -0.00344296],
        [-0.05550496,  0.05550496],
        [ 0.00080671, -0.00080671],
        ...,
        [-0.00724732,  0.00724732],
        [-0.00313857,  0.00313857],
        [-0.0447076 ,  0.0447076 ]]), shape=(105, 8, 2))

.base_values =
array([[0.5, 0.5],
       [0.5, 0.5],
       [0.5, 0.5],
       [0.5, 0.5],
       [0.5, 0.5],
       [0.5, 0.5],
       [0.5, 0.5],
       [0.5, 0.5],

```

file:///C:/Users/srish/Downloads/german_credit/html.html

```
.data =  
array([[ 33,    1,    1,    1,    0,    0, 2579,   12],  
       [ 36,    1,    3,    0,    0,    0, 5302,   18],  
       [ 33,    1,    1,    1,    1,    0,   727,   12],  
       [ 40,    1,    1,    1,    0,    0,   701,   12],  
       [ 29,    1,    2,    1,    0,    1, 1103,   12],  
       [ 25,    0,    1,    1,    1,    1, 2631,   15],  
       [ 28,    0,    2,    1,    0,    1, 4221,   30],  
       [ 20,    0,    1,    2,    0,    1, 2718,   24],  
       [ 43,    1,    2,    2,    0,    0, 4843,   12],  
       [ 35,    1,    2,    1,    0,    1, 2576,    7],  
       [ 22,    1,    2,    0,    0,    0, 3149,   24],  
       [ 24,    1,    2,    1,    1,    1, 5595,   72],  
       [ 57,    0,    2,    2,    3,    0, 3622,   30],  
       [ 20,    1,    2,    1,    1,    0,   674,   12],  
       [ 45,    0,    1,    1,    0,    2, 3049,   18],  
       [ 37,    1,    2,    1,    1,    1, 3878,   24],  
       [ 27,    0,    0,    1,    0,    0,   750,   18],  
       [ 23,    1,    2,    1,    0,    1,15672,   48],  
       [ 39,    1,    1,    2,    0,    0, 2122,   12],  
       [ 60,    1,    2,    1,    0,    0, 2246,   12],  
       [ 19,    0,    1,    2,    3,    1,   983,   12],
```

```

[ 58, 0, 1, 1, 0, 0, 385, 12],
[ 37, 0, 2, 1, 0, 1, 3612, 18],
[ 26, 1, 2, 1, 2, 1, 1158, 12],
[ 70, 1, 3, 0, 0, 1, 7308, 10],
[ 22, 1, 2, 1, 1, 1, 1514, 15],
[ 28, 1, 2, 1, 0, 1, 1068, 6],
[ 25, 0, 2, 1, 0, 0, 1835, 21],
[ 34, 0, 1, 0, 0, 1, 1837, 24],
[ 44, 0, 3, 0, 0, 1, 12579, 24],
[ 24, 0, 2, 1, 0, 0, 4272, 20],
[ 42, 1, 2, 1, 0, 0, 4153, 18],
[ 42, 1, 2, 0, 0, 1, 6288, 60],
[ 50, 0, 2, 0, 1, 1, 2671, 36],
[ 25, 0, 1, 1, 0, 2, 2327, 15],
[ 22, 1, 2, 2, 1, 1, 3092, 24],
[ 26, 0, 2, 1, 0, 1, 9960, 48],
[ 25, 1, 2, 1, 0, 0, 975, 15],
[ 48, 0, 1, 2, 0, 1, 1795, 18],
[ 27, 0, 2, 1, 0, 1, 2930, 12],
[ 49, 1, 1, 1, 0, 1, 3124, 12],
[ 63, 1, 2, 1, 0, 0, 6836, 60],
[ 55, 0, 2, 1, 0, 0, 1603, 24],
[ 30, 1, 2, 1, 0, 1, 1056, 18],
[ 23, 0, 0, 2, 2, 0, 1352, 6],
[ 37, 1, 2, 1, 0, 2, 1225, 10],
[ 29, 1, 1, 1, 0, 0, 3590, 12],
[ 32, 1, 2, 1, 0, 0, 1938, 24],
[ 20, 0, 2, 2, 0, 0, 2039, 18],
[ 22, 0, 2, 2, 0, 0, 1366, 9],
[ 27, 1, 0, 1, 3, 2, 709, 6],
[ 34, 1, 2, 1, 0, 1, 2415, 7],
[ 28, 1, 2, 1, 0, 1, 3060, 48],
[ 22, 0, 2, 1, 1, 0, 741, 12],
[ 31, 1, 2, 1, 0, 1, 1935, 24],
[ 39, 1, 3, 1, 0, 1, 11938, 24],
[ 23, 1, 2, 2, 0, 1, 1534, 12],
[ 60, 0, 3, 0, 1, 1, 14782, 60],
[ 28, 1, 3, 2, 0, 1, 9398, 36],
[ 30, 1, 2, 1, 0, 1, 7966, 26],
[ 23, 0, 1, 1, 1, 1, 836, 12],
[ 24, 1, 2, 1, 0, 2, 3016, 12],
[ 41, 1, 2, 1, 1, 1, 2503, 30],
[ 30, 1, 3, 1, 0, 0, 1823, 24],
[ 29, 0, 2, 1, 3, 0, 3357, 21],
[ 65, 1, 2, 1, 0, 0, 571, 21],
[ 31, 1, 1, 1, 1, 1, 9857, 36],
[ 20, 1, 2, 2, 0, 1, 585, 12],
[ 24, 1, 1, 1, 0, 0, 2214, 12],
[ 48, 0, 1, 0, 1, 2, 1240, 10],
[ 36, 1, 1, 1, 0, 1, 4241, 24],
[ 47, 1, 1, 1, 0, 1, 958, 12],
[ 29, 0, 2, 1, 0, 0, 1893, 12],
[ 29, 1, 2, 1, 0, 0, 6887, 36],
[ 31, 1, 2, 1, 1, 1, 6148, 20],
[ 53, 1, 2, 0, 0, 0, 4870, 24],
[ 25, 1, 2, 1, 0, 2, 5152, 24],
[ 28, 0, 2, 1, 1, 1, 2278, 18],
[ 23, 0, 3, 1, 0, 2, 1961, 18],
[ 65, 1, 2, 0, 0, 0, 2600, 18],
[ 28, 1, 2, 1, 1, 2, 1323, 6],

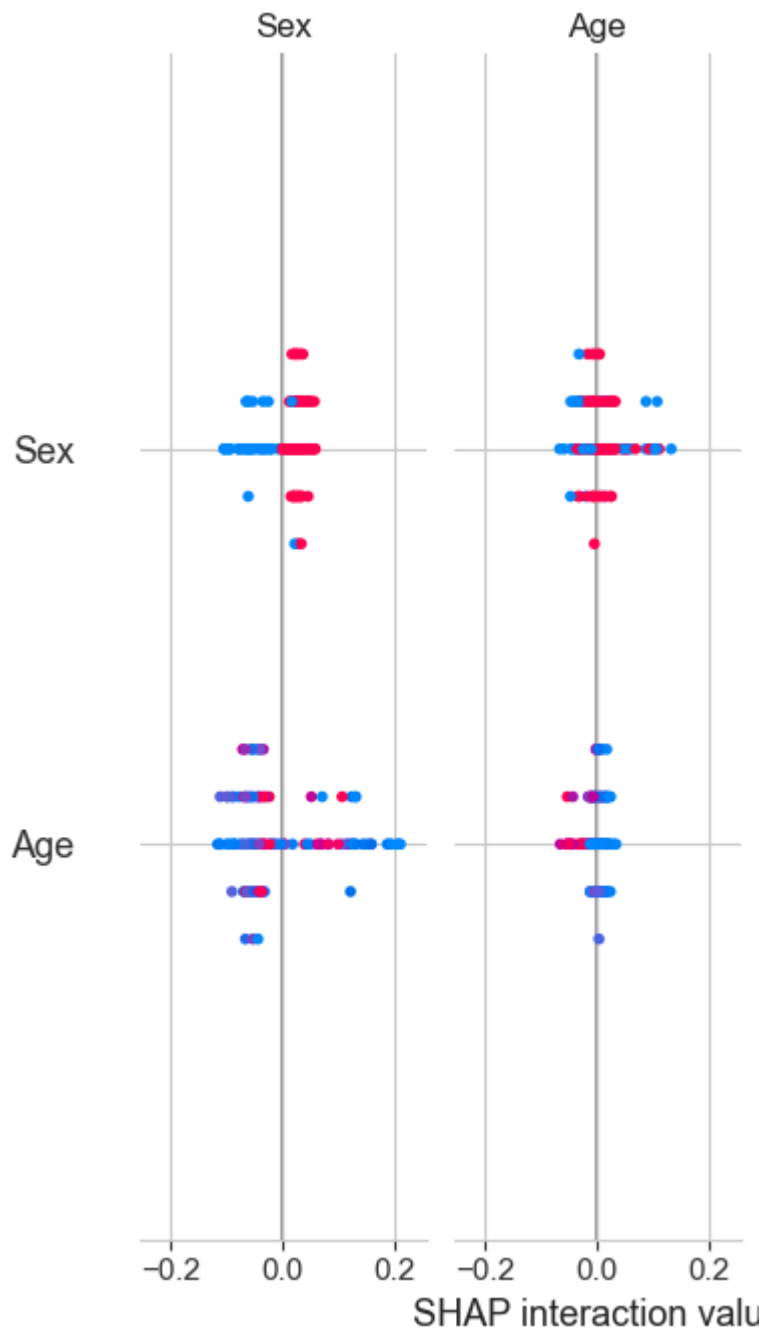
```

```
[ 44, 1, 3, 2, 0, 0, 3384, 6],
[ 39, 1, 3, 0, 0, 0, 3249, 36],
[ 44, 0, 1, 1, 0, 1, 760, 8],
[ 37, 1, 2, 2, 0, 0, 3676, 6],
[ 30, 1, 3, 1, 0, 0, 3959, 36],
[ 23, 0, 2, 2, 0, 0, 2511, 15],
[ 32, 1, 2, 0, 3, 1, 1136, 9],
[ 44, 1, 2, 1, 0, 0, 1553, 18],
[ 32, 1, 2, 1, 0, 0, 4583, 30],
[ 74, 1, 0, 1, 0, 2, 1299, 6],
[ 40, 1, 1, 1, 0, 1, 1155, 12],
[ 25, 1, 2, 2, 2, 1, 3213, 18],
[ 26, 0, 1, 2, 0, 0, 3599, 21],
[ 28, 0, 2, 1, 0, 0, 3660, 24],
[ 56, 1, 2, 0, 0, 2, 4796, 42],
[ 36, 1, 2, 2, 0, 0, 7297, 60],
[ 30, 1, 2, 1, 1, 1, 3566, 48],
[ 36, 1, 2, 1, 2, 2, 2360, 15],
[ 23, 0, 1, 2, 1, 1, 841, 12],
[ 47, 1, 3, 1, 0, 1, 1209, 6],
[ 27, 1, 2, 1, 0, 1, 2820, 36],
[ 67, 0, 3, 1, 0, 1, 1199, 9],
[ 31, 1, 1, 1, 0, 0, 3108, 30],
[ 25, 1, 2, 2, 1, 1, 1264, 15]])
```

In [186... `print(shap_values.values.shape)`

(105, 8, 2)

In [187... `shap.summary_plot(shap_values, x_test)`



```
In [188... print(x_test.shape)
```

```
(105, 8)
```

```
In [189... print(type(best_et))
```

```
<class 'sklearn.ensemble._forest.ExtraTreesClassifier'>
```

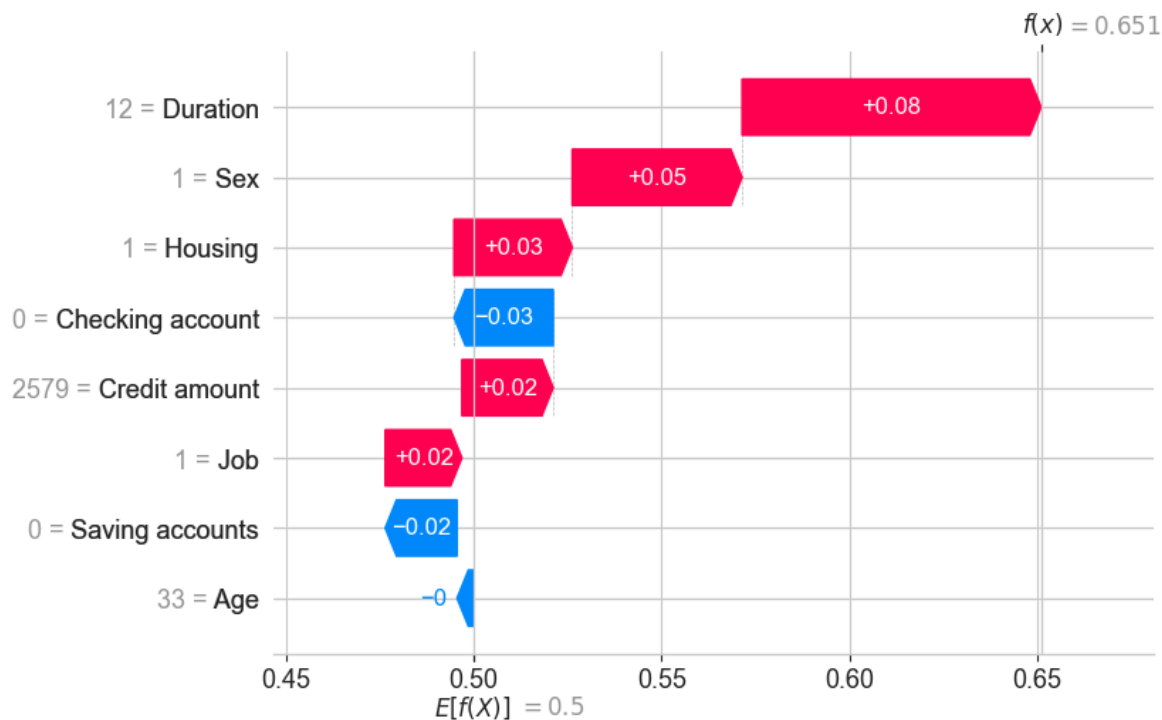
```
In [190... print(type(shap_values))
```

```
<class 'shap._explanation.Explanation'>
```

```
In [191... print(shap_values.values.shape)
```

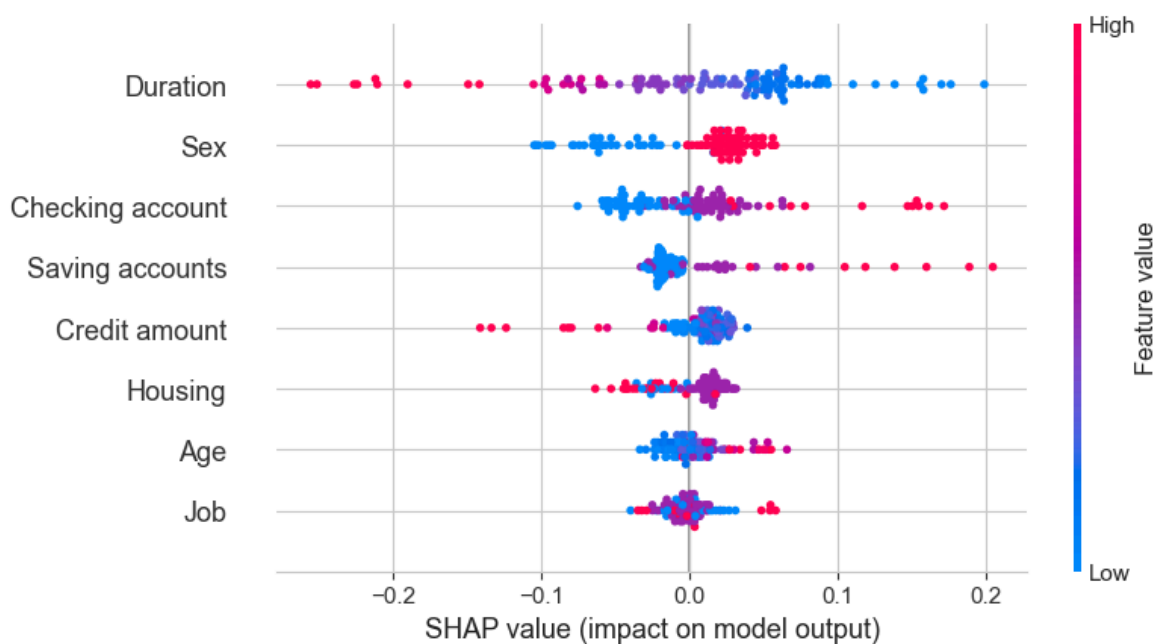
```
(105, 8, 2)
```

```
In [192... shap.plots.waterfall(shap_values[0, :, 1])
```



In []: *# duration is the most influential feature for 1st observation in class 1*

In [193... `shap.plots.beeswarm(shap_values[:, :, 1])`



In []: *#Based on this SHAP summary:*

*# Duration is the strongest predictor in the model.
 # Sex is also an influential feature.
 # Checking account and Saving accounts contribute meaningfully.
 # Age and Job have comparatively little influence.*

In [194... `shap.plots.beeswarm(shap_values[:, :, 0])`

